SentinelVision AI

Volume 01

Product Requirements Document

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> **SentinelVision AI — Enterprise Documentation Suite**

> Status legend: Implemented | Planned | Future

1. Document Control

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2. Executive Summary

SentinelVision AI transforms CCTV infrastructure into an intelligent security and activity monitoring platform using hybrid deployment: AI inference at the customer edge, centralized control plane for operations, reporting, and governance.

2.1 Current MVP (M0/M1) — Implemented

- Multi-tenant Django REST API with JWT authentication
- Organization, site, zone, camera, edge device registry
- Simulated edge person-detection event ingestion
- Dashboard summary, camera inventory, event review, alert acknowledgement
- Docker Compose local stack and basic CI

2.2 Target Platform — Planned/Future

- Production YOLO, tracking, InsightFace, threat detection modules
- Visitor management, attendance, reports, notifications
- Object storage, Celery jobs, Kubernetes production deployment
- Enterprise SSO, audit automation, biometric consent workflows

3. Business Goals

ID	Goal	KPI	Target
BR-01	Reduce incident response time	Mean time to acknowledge critical alert	< 2 min
BR-02	Improve threat detection quality	Intrusion/weapon true positive rate	>= 92%
BR-03	Automate attendance	Manual reconciliation hours/week	-80%
BR-04	Visitor compliance	Pre-registered check-in accuracy	>= 98%

ID	Goal	KPI	Target
BR-05	Platform reliability	Control plane uptime	99.9%
BR-06	Tenant growth	Active sites per customer	>= 3
BR-07	Privacy compliance	Biometric consent coverage	100%

4. Product Vision

Every camera becomes an intelligent sensor protecting people, assets, and operations with privacy-respecting AI that security teams can trust.

5. Product Scope

5.1 In Scope

Area	Status
Multi-tenant org/site/zone management	Implemented
Camera and edge device registry	Implemented
Event ingestion and review queue	Implemented
Alert rules and acknowledgement	Partial — rules create alerts; notifications Planned
Web dashboard operations UI	Implemented (core screens)
Simulated edge inference	Implemented
Face recognition and attendance	Planned
Weapon/fire/smoke detection	Future
Native mobile apps	Future

5.2 Out of Scope (MVP)

- Full VMS/NVR replacement
- Continuous cloud video archive
- Automated law enforcement dispatch
- Custom model training portal (Future)

6. User Personas

Persona	Role	Primary Goals
Sarah	Security Manager	Threat response, incident audit
Ahmed	Facility Admin	Camera config, uptime, user admin
Priya	HR Manager	Attendance, visitor compliance
James	Edge Technician	Edge install, diagnostics

Persona	Role	Primary Goals
Maria	Executive	Risk dashboards, compliance

7. User Stories

ID	Story	Priority	Status
US-01	Real-time intrusion alerts	P0	Planned
US-02	Review events with evidence	P0	Partial
US-03	Unknown person in restricted zone	P0	Planned
US-04	Weapon/fire/smoke severity alerts	P0	Future
US-05	Register cameras to sites/zones	P0	Implemented
US-06	RBAC for configuration changes	P0	Implemented
US-07	Edge device health monitoring	P1	Implemented
US-08	Automated attendance	P1	Planned
US-09	Visitor pre-registration	P1	Planned
US-10	Biometric consent before enrollment	P0	Planned
US-11	Weekly incident reports	P1	Planned
US-12	Immutable audit logs	P0	Partial — model exists, auto logging Planned

8. Functional Requirements

8.1 Camera Management (FR-CAM)

ID	Requirement	Status
FR-CAM-01	Register camera with RTSP URL, site, zone	Implemented
FR-CAM-02	ONVIF discovery	Planned
FR-CAM-03	Camera online/offline status from edge heartbeat	Implemented
FR-CAM-04	Assign AI modules per camera	Partial — field exists
FR-CAM-05	Configure detection zone polygons	Implemented (data model)

8.2 Dashboard (FR-DASH)

ID	Requirement	Status
FR-DASH-01	Overview metrics	Implemented
FR-DASH-02	Event review queue with filters	Partial
FR-DASH-03	Live camera status grid	Planned
FR-DASH-04	Role-based navigation	Partial

8.3 Alerts & Notifications (FR-ALERT)

ID	Requirement	Status
FR-ALERT-01	Configurable alert rules	Implemented
FR-ALERT-02	Severity levels info/warning/critical	Implemented
FR-ALERT-03	Email/SMS/push notifications	Planned
FR-ALERT-04	Acknowledgement workflow	Implemented

8.4 Reports (FR-RPT)

ID	Requirement	Status
FR-RPT-01	Incident summary export PDF/CSV	Planned
FR-RPT-02	Attendance report	Planned
FR-RPT-03	Visitor log report	Planned
FR-RPT-04	Audit log export	Planned

8.5 Mobile (FR-MOB)

ID	Requirement	Status
FR-MOB-01	Responsive web for alerts	Implemented
FR-MOB-02	PWA push notifications	Planned
FR-MOB-03	Native iOS/Android apps	Future

9. Non-Functional Requirements

ID	Category	Requirement	Target	Status
NFR-01	Performance	Event ingest latency p95	< 3s	Planned measurement
NFR-02	Performance	Dashboard load	< 2s	Planned measurement
NFR-03	Scalability	Cameras per edge node	16	Target
NFR-04	Availability	Control plane uptime	99.9%	Future

ID	Category	Requirement	Target	Status
NFR-05	Security	TLS in transit	1.2+	Planned production
NFR-06	Security	AES at rest	Required	Planned
NFR-07	Privacy	Biometric consent mandatory	100%	Planned
NFR-08	Retention	Event metadata default	90 days	Planned enforcement
NFR-09	Accessibility	WCAG 2.1 AA	Target	Future

10. AI Modules Overview

Module	Status	Phase
Multi-person tracking	Planned	M2
Face recognition	Planned	M3
Unknown person detection	Planned	M3
Intrusion detection	Planned	M2
Activity recognition	Future	M4
Fire/smoke detection	Future	M4
Weapon detection	Future	M4
Attendance	Planned	M3
Visitor management	Planned	M3
Vehicle detection	Future	M5
License plate recognition	Future	M5

11. Acceptance Criteria

11.1 M0/M1 Foundation — Implemented

- JWT login and tenant-scoped APIs operational
- Camera/site CRUD with RBAC enforced on writes
- Edge heartbeat and event ingest functional
- Dashboard displays summary, cameras, events, alerts
- Docker Compose stack starts successfully
- CI runs backend, edge, frontend, and documentation builds

11.2 M2 Core AI — Planned

- One RTSP stream connected at edge
- YOLO person detection with tracking IDs
- Zone-based intrusion events generated
- Operator confirms event and acknowledges alert end-to-end

12. Known Gaps (Remediation Required)

Gap	Impact	Remediation
Audit auto-logging not wired	Compliance gap	Implement log_audit callers
Global edge_event_id lookup	Tenant isolation risk	Scope idempotency per organization
Cross-site zone assignment possible	Data integrity risk	Add serializer validation
Fallback EDGE_API_KEY without device binding	Security risk	Remove or bind to bootstrap device
Notification dispatch absent	Operational gap	Add Celery notification service

13. Roadmap

Phase	Timeline	Deliverables
M0/M1	Complete	Foundation, simulated pipeline
M2	Weeks 4-10	RTSP, YOLO, intrusion, alerts
M3	Weeks 11-14	Face, attendance, visitors
M4	Weeks 15-18	Threat modules, reports
M5	Weeks 19-24	Enterprise HA, SSO, analytics

14. Traceability

See `docs/traceability-matrix.csv` for BR/FR/NFR to UC/API/DB/TEST mappings.